

MOD 312 Course Outline

MOD 312 — Integrated Macro-Sector Modelling

Course Outline

School	03 Economic Modelling
Level	300 (practitioner / intermediate to advanced)
Hours	39.9998 (indicative band 36–45)
Shape	5 parts · 13 sessions · 7 laboratories · 4 assignments · 1 examination · 1 final project
Delivery	Instructor-led — online, blended or intensive in person
Prerequisites	The rest of the school. MOD 303 and 305 for the solver, MOD 306 for CDCGE, MOD 308 for the multi-regional table, MOD 309 for the distributional link and MOD 310 for the diagnostic frame. This course integrates rather than introduces.
Assessment	Pass 60 %, with 60 % laboratory competency
Version	1.0 · reviewed 2026-08-27 · 12-month cycle

What this course is

Every other course in this school assumes a model that solves. This one asks whether it does. A 64-run experiment crosses four regions, two sectoral resolutions, six policy shocks and two start strategies, and records for each whether the solver converged. Fifty-one did. The course reads that grid as what it is — an experiment about a model rather than about an economy — and then assembles the whole school’s material into a single integrated assessment, stating at every step which pieces exist and which do not.

The capstone’s subject is a sixty-four-run solver grid, and its first discipline is admitting what that is: an experiment about a model, whose row labels being country names does not make it a fact about any economy. Two of its results are negative and both are reported as findings — no output cost exists for any scenario anywhere in this school, and the grid’s design cannot support a recommendation about start strategy because the comparison is confounded. The integrated assessment of a carbon price stops at four of eight steps, with the other four named and a specific fix given for each. All 51 registered findings are in the appendix, and the asymmetry across them is itself a session.

The rule the course is built on

A grid is an experiment about a model. Everything downstream follows from holding that line: an outcome column that is empty is not an outcome of zero, a comparison whose arms differ in two things at once settles neither, and a chain of models that stops halfway

is reported as stopping halfway with the remaining steps specified. Naming what did not happen is the deliverable, not the apology.

Who it is for

Modellers who own more than one model and have to make them agree, and the reviewer or commissioner who has to judge whether an integrated assessment's chain of models actually closed.

Structure

Part	Title	Sessions	Laboratory	Hours
P1	What integration means, and the grid	01–02	01	4.8
P2	Convergence, replication and resolution	03–05	02–03	8.2
P3	Shocks, warm and cold starts, reading the grid	06–08	—	4.2
P4	The focus economy, and two negative results	09–11	04–05	8.2
P5	Assembling the school, and reporting	12–13	06–07	6.8
	Assignment and assessment time			8.0
	Total	13	7	40.0

The 13 sessions and 7 laboratories, in teaching order, and the subject matter of each:

P1 What integration means, and the grid — Sessions 01–02 · 4.8 h · Laboratory 01

- **01 What Integration Means** — The models this school carries; What each answers and what it cannot; The joins: SAM to CGE, CGE to micro, MRIO to structure; Soft links, hard links and iterated links; Which of this school's joins actually complete; What integration is not
- **02 The Solver Grid** — Why map convergence at all; The design: regions, resolutions, shocks, starts; What each run records; The grid's own declared purpose; Reading a grid as an experiment; What a solve rate is and is not
- *Lab 01 Read the Grid* — What experiment is this, and what did it find?

P2 Convergence, replication and resolution — Sessions 03–05 · 8.2 h · Laboratory 02–03

- **03 Convergence** — Mixed complementarity, restated; `LOCALLY_SOLVED` and what the qualifier does; `SLOW_PROGRESS`: the solver's criterion, not the clock; Time to failure against time to success; Why a failed run is information; What a modeller does with a non-convergence
- **04 The Benchmark Replication** — What replication tests; The maximum relative deviation; Reading a deviation of 1e-10; Replication succeeding where scenarios fail; What replication does not establish; The eight benchmark runs in this grid
- **05 Resolution** — Nine sectors against thirty; The variable count, and what drives it; Runtime and solve rate against resolution; How much the reported outcome moves; The aggregation bias question; Choosing a resolution honestly
- *Lab 02 Where the Failures Are* — Is failure spread evenly, or concentrated?
- *Lab 03 Replication and Resolution* — Does the model reproduce its base year, and what does detail cost?

P3 Shocks, warm and cold starts, reading the grid — Sessions 06–08 · 4.2 h

- **06 Shocks** — The six shocks in this grid; Carbon pricing at three levels; A tax rise and a tariff; Solve rate by shock; The emissions response, read carefully; Monotonicity as a sanity check
- **07 Warm and Cold Starts** — What a warm start is; Why it usually helps and sometimes does not; The path dependence question; Designing a test of start strategy; What this grid's design permits; And what it does not
- **08 Reading the Grid** — The headline solve rate; Marginal rates by every dimension; Where the failures concentrate; Interaction, and why a grid shows it; What a 79.7 per cent solve rate means for a work plan; Reporting a grid to a commissioner

P4 The focus economy, and two negative results — Sessions 09–11 · 8.2 h · Laboratory 04–05

- **09 The Focus Economy** — Six of sixteen; Ten of thirteen failures; Every Utopian carbon run; And the benchmark replicates cleanly; What that combination does and does not imply; The consequence for MOD 306, 308, 309 and 310
- **10 An Outcome That Is Not There** — Two outcome columns, one populated; Forty-three runs, forty-three emissions figures, zero cost figures; What a carbon price costs, and why nobody here can say; One-sided evidence and the direction of its bias; Screening an outcome column before quoting a scenario; Reporting a response with its missing counterpart
- **11 A Confounded Comparison** — The design, laid out as a table; Four of six shocks at one start only; The aggregate comparison; The balanced comparison; Why they point in opposite directions; What sixteen runs an arm can support
- *Lab 04 An Outcome That Is Not There* — What does a carbon tax cost?
- *Lab 05 A Confounded Comparison* — Do warm starts help?

P5 Assembling the school, and reporting — Sessions 12–13 · 6.8 h · Laboratory 06–07

- **12 Assembling the School** — The full inventory; Which joins complete and which do not; The twenty-nine school gaps, by course; What the school can answer end to end; What it cannot; The integrated assessment, attempted
- **13 Reporting Integrated Work** — What an integrated note must state; Availability before results; How integrated modelling is oversold; The chain-of-custody problem; Reporting a link that does not complete; The integrated modelling note template
- *Lab 06 The School Inventory* — What does this school actually contain?
- *Lab 07 The Integrated Note* — What can this school answer end to end?

What a graduate can do

1. Explain what an integrated macro-sector modelling system is and what it is for.
2. Describe a solver grid and say what question it answers.
3. Read a convergence status and say what it does and does not mean.
4. Compute a solve rate and say what it can support.
5. Distinguish a benchmark replication from a counterfactual run.
6. Interpret a maximum relative deviation.
7. Say what sectoral resolution costs and what it buys.
8. Screen an outcome column for emptiness before quoting a scenario.
9. Recognise a confounded comparison in an unbalanced design.
10. Assemble the school's models into one integrated assessment.
11. State which links between models exist and which do not.
12. Write an integrated modelling note that a commissioner can act on.

The formal learning outcomes, and the map from each to the assessment that tests it, are in 12_LMS/learning_outcomes.md.

Assessment

Component	Weight	What it is
Knowledge quizzes	—	A bank of 104 items, used for formative practice and drawn on by the theory paper, which carries the marks
Laboratory exercises	35 %	7 laboratories, equally weighted. A laboratory is not complete until its validation checks have been read and its named deliverable produced: an irreproducible result is no result
Applied assignments	20 %	Read the experiment; Screen the outcomes; A confounded comparison; The integrated note
Theory examination	20 %	Closed-book, drawn from the quiz bank and the review questions

Component	Weight	What it is
Final project	25 %	A complete integrated modelling assessment — of a policy question of the learner’s choosing, using this school’s models or a system the learner is responsible for — delivered as an eight-section note plus the screening scripts that produced it

Pass 60 % overall and 60 % laboratory competency. Merit 70 %. Distinction 80 %. One resubmission of a single failed component within eight weeks. A laboratory is not complete until every validation check has been read and the provenance of every figure recorded: an irreproducible result is no result.

Data and access

No restricted data and no household record. The course states no fact about any economy, reports no output cost for any scenario and makes no recommendation about start strategy, in each case because the evidence cannot carry it. It repairs nothing: no re-run with a populated outcome column, no rebalanced design, and no convergence work on the region that fails.

Materials and set-up

A 120-page Technical Manual, 14 lecture decks of 215 slides, 7 laboratory decks of 91 slides, a Laboratory Manual, Instructor Guide and Learner Workbook, a 104-item quiz bank, 5 assessment files, 39 datasets, 4 code exercises, 3 case studies and the LMS package. The README lists every file, and COURSE_BUILD_REPORT.md records how each was produced.

The grid ships with the course as a table and needs no model to read. Learners who want to reproduce a cell need CDCGE at D:/EDIP/models/CDCGE, the PATH licence and Julia with `-pkgimages=no`; the Laboratory Manual says which cells are cheap to reproduce and which take an hour.

The 51 findings registered against this school, and the bearing each has on this course, are carried in the Technical Manual’s gap appendix and listed in `COURSE_VERSION.json`.

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